

Certified analytical bounds for hypersonic reachable sets

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Abstract

This is a proposal in nonlinear analysis and dynamical systems. The object is a singularly perturbed family of differential inclusions in which compactness fails along the time axis: as the parameter vanishes, the motion carrying the important state transitions occupies a vanishing time, so it disappears from the ordinary limits of the family while still deciding where the trajectories go. The natural remedy is a *profile decomposition* — a low-dimensional *skeleton*, finitely many concentrating *bubbles*, a *residual* bounded in a stated metric — and the question is whether that is a theorem about solution sets or only a description of them.

We state it as a hierarchy. *Theorem A*: every sequence of trajectories splits, along a subsequence, into a skeleton, finitely many rescaled profiles above any fixed size, and a remainder. *Theorem B*: the profiles induce a set-valued jump map. *Theorem C*: the solution set of the family lies in that of the skeleton-and-jump system, thickened by an explicit ball. This document proves Theorem C, with every constant written down, for a model whose potential is asymmetric in the one way that produces concentration.

It does not prove Theorem A, and the relation wants stating exactly: the theorem proved here establishes no decomposition and its proof uses none. What it does is twofold. It is the quantitative estimate any decomposition must reproduce — including the profile count and the amplification they compound, which a Lyapunov functional supplies *a priori* where the classical argument obtains them by contradiction and without a constant. And it shows the concentration structure can be reached without the decomposition machinery at all, by an exact integral identity and a budget. Whether that machinery buys anything the elementary route does not is the first question of the programme.

Hypersonic entry supplies the problem and is the test the result is measured against; it is not the subject. Only the theorem of [Section 4](#) is claimed as proved.

1 The question, reduced

The object we want enclosed is the solution set of a family of differential inclusions: every state reachable under every admissible control and every admissible selection. Two things make that hard. Its dimension puts the exact methods out of reach; and the bound must come with a proof, which is to be a pair of functions whose superlevel set contains the solution set, exhibited and checked rather than produced by a solver ([Section 6](#)). This document is narrower than the programme and prior to it: it takes the analytical difficulty the programme turns on, strips it of the application, and proves a theorem about it.

The difficulty is a singular limit that defeats the ordinary compactness arguments. For atmospheric entry the parameter is $\varepsilon = H_s/R_e$, the ratio of scale height to planetary radius, whose square root is the ratio of the oscillation period about equilibrium to the slow time of the flight. As $\varepsilon \rightarrow 0$ the motion separates into arcs where the forcing is not felt and *passes* where it is felt strongly for a vanishing time. A pass becomes instantaneous and keeps its effect, so in the limit a trajectory is a sequence of arcs joined by jumps — yet the natural limiting objects do not see the jumps, since they weigh an interval by its length and a pass has none. A solution set is a support, not an integral, so what has no mass still counts.

The language for this is a profile decomposition, obtained in the theory of nonlinear dispersive equations by iterating the dichotomy of the concentration-compactness principle [\[1, 2\]](#). It delivers structure — which pieces can occur, and that nothing else can, by contradiction — and no constant: the authors of the wave-equation theory are explicit that the uniform bound following from their decomposition carries a constant they cannot describe [\[2, Corollary 2\]](#). Our proposal is that here the structure can be had *quantitatively*, because the family

carries a Lyapunov functional of bounded total variation, uniformly in the parameter. That bounds in advance the number of profiles above any fixed size, the time the motion may spend concentrating, and the accumulated growth of the comparison. The count is [Eq. \(3.14\)](#), proved below; the decomposition it would organise is not.

Remark 1.1 (What the small parameter is, and what it is not). It is tempting to write the reduced problem as a slow state driven by a fast one, $\dot{z} = g(z) + h(y)$ with $\varepsilon \dot{y} = q(y, z)$ and h bounded. That form does not contain the difficulty: if the excursion in y lasts a time of order ε and the forcing is of order one, its contribution to z is of order ε , so in the limit it leaves no trace and ordinary averaging describes everything. What produces a jump is an *asymmetric potential* together with a scaling that keeps the oscillation slow while the asymmetry sharpens. In [Section 2](#) the air supplies a wall steepening without bound below the equilibrium while gravity is uniform above it; and because ε enters only as the *width of the band* in which the air is felt, a rebound occupies a vanishing fraction of its oscillation while delivering an impulse of order one. [Lemmas 3.3](#) and [3.4](#) are those two statements. Concentration is the asymmetry of the well read at the right scale, not an assumption about the size of a force.

1.1 What this document does not prove

[Theorem 4.2](#) is Theorem C, for one model. Three statements belong to the programme and are not proved here, and each is stated in full in [Section 8](#). *Theorem A*, the decomposition itself, of which [Theorem 4.2](#) asserts nothing: what [Assumption 2.4](#) supplies is the count [Eq. \(3.14\)](#), and proving the decomposition it would organise is [objective 1](#). *The limit of the family in the weak topology*, defined in [Section 6](#) and owed by [objective 4](#). And *the passage from the weak limit back to the support*, the hardest link in the chain: granting the limit of the measures and the profiles it carries, that the limiting solution set lies in the one those profiles generate is a further implication, which the proof below avoids by getting its inclusion from trajectories directly. Which route the general case should take is open, and it is the question we would most value an advisor's judgment on. A fourth obligation is deferred rather than open: a bound in closed form must finally be *evaluated*, and a guarantee does not ordinarily survive floating point, the setting in which it does being validated interval arithmetic [[3](#), Ch. 3]. That concerns evaluating a bound rather than proving one, and we claim no computation.

1.2 Terminology

Two fields meet here and several words mean different things in each; we fix them once.

Orbit. One closed curve of [Eqs. \(2.1\) to \(2.2\)](#) at *frozen* (c, θ) and zero damping ([Lemma 2.8](#)). Not a trajectory about the planet, and not a piece of an actual trajectory, which closes no orbit: orbits serve only as the comparison objects defining P_0, D_0 in [Lemma 3.3](#). An actual trajectory is decomposed by [Definition 3.5](#), which cuts it at its crossings of $z = 0$ into *passes* ($z \leq 0$) and *arcs* ($z \geq 0$). **Selection θ .** One member of the admissible family of fields, *fixed along a trajectory* and quantified over at the end ([Remark 2.3](#)); never a function of time in what is proved here. **Profile m_θ versus density m_θ^ε .** The profile is the ε -free function of [Assumption 2.1](#), read in units of the band's thickness; the density is the profile as the motion meets it, $m_\theta^\varepsilon(z) = m_\theta(z/\varepsilon)$. Hypotheses are placed on the profile, which is why they are uniform in ε . **Lyapunov functional.** Two objects wear the name and do opposite jobs. The dissipation Θ_d of [Assumption 2.4](#) and the amplitude W are non-increasing and measure what the motion spends; every count below rests on them. The *dual pair* (Φ, Ψ) of [Section 6](#) is non-decreasing, and $\{\Psi \geq 1\}$ encloses the reachable set; that pair is the certificate, and exhibiting one is a proof whether or not it is ever evaluated. The first kind is in hand; constructing the second is [objective 4](#). **Averaging.** Always of the fast part, with a single phase ([Lemma 4.6](#)); over *several* phases, whose frequencies may cross, it is a different problem, attempted nowhere below, and is [objective 3](#). **The oscillator.** Always the motion of [Eqs. \(2.1\) to \(2.2\)](#), the altitude about the glide; the *attitude* oscillator of [objective 3](#), three orders of magnitude faster, is named in full.

2 The model and its hypotheses

Fix an integer $d \geq 1$ and a horizon $T > 0$. The state is $x = (z, v, s)$, with $z \in \mathbb{R}$ an altitude, $v \in \mathbb{R}$ its rate, and $s \in \mathbb{R}^d$ a vector of slow coordinates. The system is

$$\dot{z} = v, \tag{2.1}$$

$$\dot{v} = c m_\theta^\varepsilon(z) - 1 - \kappa(s, c, \theta) v, \tag{2.2}$$

$$\dot{s} = F(s, c, \theta) + m_\theta^\varepsilon(z) \Gamma(s, c, \theta). \tag{2.3}$$

Here $c(\cdot)$ is the control and $\theta \in \Theta$ is a *selection* — a choice of admissible field from a bounded family, fixed along each trajectory and quantified over at the end ([Remark 2.3](#)). The small parameter $\varepsilon \in (0, 1]$ appears in one place only, inside the density: the *profile* m_θ is read on the layer's own scale,

$$m_\theta^\varepsilon(z) = m_\theta(z/\varepsilon), \quad M_\theta^\varepsilon(z) = \varepsilon M_\theta(z/\varepsilon), \tag{2.4}$$

where M_θ is the antiderivative of m_θ with $M_\theta(0) = 0$, so that $(M_\theta^\varepsilon)' = m_\theta^\varepsilon$ and $M_\theta^\varepsilon(0) = 0$. The *potential* is

$$\Phi_{c,\theta}(z) = z - c M_\theta^\varepsilon(z), \quad \text{so} \quad \Phi'_{c,\theta}(z) = 1 - c m_\theta^\varepsilon(z), \quad (2.5)$$

and Eqs. (2.1) to (2.2) are a unit-mass particle in that potential with a weak damping. We write $\Phi_{c,\theta}$ without decoration, though it depends on ε : all of that dependence is the density's, and Lemma 3.3 is where it is unpacked. The *amplitude* is its energy,

$$W = \frac{1}{2}v^2 + \Phi_{c,\theta}(z). \quad (2.6)$$

The whole of Section 3 is an account of one object: the shape of this potential.

That ε sits inside the density and nowhere else is the whole of the scaling. The air is felt over a band of altitudes of thickness ε while the motion of interest is not thin: a large oscillation rises to an altitude of order one and returns, so a pass occupies a vanishing *fraction* of each oscillation without the oscillation itself becoming fast. That is the configuration in which a pass concentrates, and Lemmas 3.3 and 3.4 are the two statements that establish it — each of which also says what the other choice of scaling would have given instead.

Assumption 2.1 (H1: the wall). For each $\theta \in \Theta$ the profile $m_\theta \in C^1(\mathbb{R})$ satisfies $m_\theta > 0$ and $m'_\theta < 0$; $m_\theta(\xi) \rightarrow 0$ as $\xi \rightarrow +\infty$ with $\int_0^\infty m_\theta < \infty$; $m_\theta(\xi) \rightarrow \infty$ as $\xi \rightarrow -\infty$; and $1/c$ lies in the range of m_θ for every $c \in [c_{\min}, c_{\max}]$. We write $m_+ = \sup_\Theta m_\theta(0)$ and $M_\infty = \sup_\Theta \int_0^\infty m_\theta$, both finite and independent of ε .

The two one-sided conditions are the whole physical content: the air thins upward without bound on the integral, and thickens downward without limit. Nothing in what follows uses the shape of m_θ beyond Assumption 2.1, which is why θ may select it: an enclosure of an uncertain density is exactly a bounded family of admissible profiles. Because m_θ decreases, m_+ bounds m_θ^ε at and above the section $z = 0$ for every ε , and that is the only place a pointwise bound on the density is used below; inside the band m_θ^ε is unbounded as $\varepsilon \rightarrow 0$ and none is taken, what is used there being its *impulse*, which Assumption 2.4 bounds.

Assumption 2.2 (H2: data). $[c_{\min}, c_{\max}] \subset (0, \infty)$, Θ and $S \subset \mathbb{R}^d$ are compact. The maps $F, \Gamma: S \times [c_{\min}, c_{\max}] \times \Theta \rightarrow \mathbb{R}^d$ and κ are continuous, bounded by $F_{\max}$, $\Gamma_{\max}$ and $\kappa_{\max}$, and Lipschitz in s uniformly in (c, θ) with constants κ_F, κ_Γ . Moreover $\kappa > 0$. The selection θ is a fixed element of Θ along each trajectory; only $c(\cdot)$ varies with time.

Remark 2.3 (Why the selection is fixed, and what that costs). The reachable set below is the union over $\theta \in \Theta$ of the sets reached at fixed θ : the uncertainty is quantified over, not allowed to vary *during* a trajectory. The reason wants stating, since it is where a hidden hypothesis would otherwise sit. The potential Eq. (2.5) depends on θ , so a time-varying θ would move the well beneath the amplitude Eq. (2.6) and Eq. (3.1) would acquire a term in $\partial_\theta M_\theta^\varepsilon(z)\dot{\theta}$ — which for a merely measurable $\theta(\cdot)$ one cannot bound, or in general even write. Every estimate descending from the amplitude would go with it.

The budget does not depend on this. Assumption 2.4 asks for one functional decreasing for *all* (s, c, θ) at once, so Eq. (5.1) holds for an arbitrary measurable $\theta(\cdot)$ as it stands. The division is clean: the slow budget survives a time-varying selection, the fast amplitude law does not.

The cost is real and runs the unsafe way. The set reached when the atmosphere and the coefficients may vary in time is *larger* than the union taken here, so Theorem 4.2 bounds the union and not that larger set. Enlarging the class to selections of bounded variation in altitude — what the physics supplies, the atmosphere being smooth in space and the mass lost monotonically — is objective 2, an enlargement of the enclosure rather than a refinement of its constants.

Assumption 2.4 (H3: a Lyapunov functional). There is $\Theta_d \in C^1(\mathbb{R}^d)$ and a constant $c_0 > 0$ with

$$\nabla \Theta_d(s) \cdot F(s, c, \theta) \leq 0, \quad \nabla \Theta_d(s) \cdot \Gamma(s, c, \theta) \leq -c_0,$$

for all (s, c, θ) . Write $M = (\max_S \Theta_d - \min_S \Theta_d)/c_0$.

A Lyapunov functional for the slow motion, non-increasing along every trajectory for every control and every selection. Bounding its total variation over the compact corridor converts monotonicity into a count: by Eq. (5.1) it bounds $\int m_\theta^\varepsilon d\tau$ uniformly in ε , and every count in Section 5 descends from that one inequality. In the application it is the specific mechanical energy, falling at the drag power because neither the lift nor the Coriolis force does work.

Assumption 2.5 (H4: control). $c(\cdot)$ is absolutely continuous with values in $[c_{\min}, c_{\max}]$ and $|\dot{c}| \leq \dot{c}_{\max}$.

Assumption 2.6 (H5: corridor). Every trajectory remains on $[0, T]$ in a compact corridor $\mathcal{X} = \{|z| \leq z_{\max}, |v| \leq v_{\max}, s \in S\}$, and $x(0) \in \mathcal{X}_0 \subseteq \mathcal{X}$. Write $W_{\max} = \frac{1}{2}v_{\max}^2 + z_{\max}$ for the largest amplitude it admits.

The corridor is a fixed box, which the scaling of Eq. (2.4) is what permits: the oscillations of interest have amplitude of order one, so bounding them costs nothing that shrinks. Only the antiderivative needs care, being unbounded below as $\varepsilon \rightarrow 0$ at fixed $z < 0$ — though not where a trajectory can reach. Wherever $\Phi_{c,\theta}(z) \leq W_{\max}$ we have $-c M_\theta^\varepsilon(z) = \Phi_{c,\theta}(z) - z \leq W_{\max} + z_{\max}$, and above the section $|M_\theta^\varepsilon| \leq \varepsilon M_\infty$, so

$$M_* := M_\infty + \frac{W_{\max} + z_{\max}}{c_{\min}} \quad (2.7)$$

bounds $|M_\theta^\varepsilon|$ on the reachable part of the corridor, uniformly in ε .

Remark 2.7 (What is uniform, and what the hypotheses are not). Every constant named above is a bound over $S \times [c_{\min}, c_{\max}] \times \Theta$. No constant below depends on the particular control, the particular selection, or the particular initial point: that is the sense in which the result is uniform over the admissible family, and it is why an enclosure of an uncertain field is the natural setting rather than a complication. Note also what is absent. There is no transversality condition, no non-grazing condition, and no assumption that a pass is a single dip — the last is Lemma 2.8, a consequence of $m'_\theta < 0$. Every hypothesis is a condition on the data, checkable before any trajectory is computed.

Lemma 2.8 (One well, and one dip per period). *Under Assumption 2.1 the potential $\Phi_{c,\theta}$ is strictly convex, with*

$$\Phi_{c,\theta}''(z) = -\frac{c}{\varepsilon} m'_\theta(z/\varepsilon) > 0, \quad (2.8)$$

and attains its minimum at the unique $z_(c, \theta) = \varepsilon \xi_*$, where ξ_* is the unique point with $cm_\theta(\xi_*) = 1$: the glide sits inside the band, at height of order ε . Consequently every sublevel set $\{\Phi_{c,\theta} \leq W\}$ is a bounded interval $[z_-(W), z_+(W)]$, and at frozen (c, θ) every bounded motion of Eqs. (2.1) to (2.2) with $\kappa = 0$ is a single closed orbit — one closed curve in the (z, v) plane, traversed periodically — on which z attains exactly one minimum and one maximum.*

Proof. Differentiating Eq. (2.5) twice with Eq. (2.4) gives Eq. (2.8), positive because $c, \varepsilon > 0$ and $m'_\theta < 0$. A strictly convex C^2 function has at most one stationary point, and $\Phi_{c,\theta}' = 1 - cm_\theta^\varepsilon$ changes sign exactly once since m_θ decreases strictly from $+\infty$ to 0 with $1/c$ in its range; so z_* exists and is unique. Convexity makes each sublevel set an interval, bounded because $\Phi_{c,\theta} \rightarrow +\infty$ at both ends. At frozen (c, θ) and $\kappa = 0$, Eq. (2.6) is constant, so the orbit is the level curve $\{\frac{1}{2}v^2 = W - \Phi_{c,\theta}(z)\}$ — a single closed curve meeting $v = 0$ exactly at the interval's endpoints, which are the one minimum and one maximum of z . $\square$

Lemma 2.8 is worth noting for what it is not: a hypothesis. That a pass is a single dip rather than a train of them is a property of the motion, and here it costs one sign on the density rather than a condition imposed on solutions.

3 Three regimes of one oscillator

Glide, phugoid and skip are not three systems but three energies in the single well of Lemma 2.8. This section says which is which. The statement that organises everything is that ε sets no shape, only a clock — and that the clock is read at amplitude W/ε , which is what makes a pass concentrate rather than average away.

Lemma 3.1 (The amplitude law). *Along every trajectory of Eqs. (2.1) to (2.3),*

$$\dot{W} = -\kappa(s, c, \theta) v^2 - M_\theta^\varepsilon(z) \dot{c}, \quad (3.1)$$

whatever the profile m_θ and whatever ε ; the only time-dependent datum the potential carries is c , which is why Eq. (3.1) closes (Remark 2.3). Consequently, with M_ the uniform bound Eq. (2.7),*

$$W(\tau) \leq W(0) + M_* \dot{c}_{\max} \tau \quad \text{for } \tau \in [0, T]. \quad (3.2)$$

Proof. Differentiate Eq. (2.6) along Eqs. (2.1) to (2.2):

$$\dot{W} = v\dot{v} + \Phi_{c,\theta}'(z) \dot{z} + \frac{\partial \Phi_{c,\theta}}{\partial c} \dot{c} = v(cm_\theta^\varepsilon - 1) - \kappa v^2 + (1 - cm_\theta^\varepsilon)v - M_\theta^\varepsilon(z) \dot{c},$$

where $\partial \Phi_{c,\theta} / \partial c = -M_\theta^\varepsilon(z)$ by Eq. (2.5). The two terms carrying cm_θ^ε cancel identically, which is Eq. (3.1); the cancellation uses only that the same cm_θ^ε appears in Eq. (2.2) and in $\Phi_{c,\theta}'$. Since $\kappa > 0$, Eq. (3.1) gives $\dot{W} \leq M_* \dot{c}_{\max}$, and Eq. (3.2) follows by integration. $\square$

So W is a Lyapunov functional for the fast motion, decaying except where the control pumps it, and Assumption 2.5 makes Eq. (3.2) an *a priori* bound rather than a hypothesis about solutions. With the functional of Assumption 2.4 — one for the fast motion, one for the slow — these are the two monotone quantities every count below rests on.

3.1 The mean density, and the time the air is felt

Recall from [Section 1.2](#) that an *orbit* here is a closed curve of the *frozen, undamped* system, never a piece of an actual trajectory. Two exact identities do much of the quantitative work, and they have different standings: the first holds on such an orbit and comes from periodicity, the second holds along the trajectory as it is and comes from the budget alone.

Lemma 3.2 (The mean density of an orbit). *Let $\langle \cdot \rangle$ denote the time average over one closed orbit of [Eqs. \(2.1\) to \(2.2\)](#) at frozen (c, θ) and $\kappa = 0$. Then*

$$\langle m_\theta^\varepsilon \rangle = \frac{1}{c}, \quad (3.3)$$

for every amplitude W , every profile satisfying [Assumption 2.1](#) and every ε ; the impulse of such an orbit equals its period divided by c . Separately, and along the trajectory as it is rather than along any frozen orbit, [Eq. \(5.1\)](#) bounds the time the motion may spend where the air is appreciable: for every $\mu > 0$,

$$|\{\tau \in [0, T] : m_\theta^\varepsilon(z(\tau)) \geq \mu\}| \leq \frac{M}{\mu}. \quad (3.4)$$

Proof. Integrate [Eq. \(2.2\)](#) over one period at $\kappa = 0$. The left side is $[v] = 0$ because v is periodic, so $\int (c m_\theta^\varepsilon - 1) d\tau = 0$, that is $c \int m_\theta^\varepsilon d\tau = \int d\tau$, which is [Eq. \(3.3\)](#) on dividing by the period. The impulse of the orbit is $\int m_\theta^\varepsilon d\tau = \langle m_\theta^\varepsilon \rangle \times (\text{period}) = (\text{period})/c$. For [Eq. \(3.4\)](#), the set where $m_\theta^\varepsilon \geq \mu$ contributes at least μ times its measure to $\int_0^T m_\theta^\varepsilon d\tau$, which [Eq. \(5.1\)](#) bounds by M ; this uses no periodicity and holds for an arbitrary measurable selection. $\square$

[Equation \(3.3\)](#) is the reason the skeleton is the glide: whatever the oscillation does, its time-averaged density equals the density at the equilibrium, so the glide is not merely the bottom of the well but the *mean* of every oscillation about it. Since $\langle m_\theta^\varepsilon \rangle$ does not depend on W , neither does the averaged slow motion, and the amplitude is an output of the reduced description rather than a driver of it. Note the standing of the identity: it concerns the frozen, undamped system, and the proof nowhere assumes the trajectory follows such an orbit. It supplies the *form* of the averaged field in [Definition 4.1](#), and [Lemma 4.6](#) estimates the error of using that field without any appeal to periodicity.

3.2 What ε does, and what it does not

Lemma 3.3 (ε is a clock, and the amplitude it is read against). *Let $\xi = z/\varepsilon$ and $u = v/\sqrt{\varepsilon}$, and let $\Phi_{c,\theta}^0(\xi) = \xi - c M_\theta(\xi)$ be the ε -free potential of the profile. At frozen (c, θ) and $\kappa = 0$ the orbit through amplitude W is, in the (ξ, u) plane, the level set $\{\frac{1}{2}u^2 = \Omega - \Phi_{c,\theta}^0(\xi)\}$ at*

$$\Omega = W/\varepsilon, \quad (3.5)$$

which does not involve ε ; and $d\tau = \sqrt{\varepsilon} d\xi/u$, so every time measured along the orbit is $\sqrt{\varepsilon}$ times a quantity depending on (Ω, c, θ) alone. In particular the period and the time below the section $z = 0$ are

$$P = \sqrt{\varepsilon} P_0(\Omega, c, \theta), \quad P_{\text{dip}} = \sqrt{\varepsilon} D_0(\Omega, c, \theta). \quad (3.6)$$

Proof. Substituting $z = \varepsilon\xi$ in [Eq. \(2.4\)](#) gives $m_\theta^\varepsilon(z) = m_\theta(\xi)$ and $M_\theta^\varepsilon(z) = \varepsilon M_\theta(\xi)$, so $\Phi_{c,\theta}(z) = \varepsilon \Phi_{c,\theta}^0(\xi)$. Then $W - \Phi_{c,\theta}(z) = \varepsilon(\Omega - \Phi_{c,\theta}^0(\xi))$ and $\frac{1}{2}v^2 = \varepsilon \cdot \frac{1}{2}u^2$, which is the stated level set. [Equation \(2.1\)](#) gives $d\tau = dz/v = \varepsilon d\xi/(\sqrt{\varepsilon}u) = \sqrt{\varepsilon} d\xi/u$, so every elapsed time is $\sqrt{\varepsilon}$ times an integral over ξ of a function of (Ω, c, θ) . Taking it over the whole orbit and over its part with $\xi \leq 0$ gives [Eq. \(3.6\)](#). $\square$

So ε still sets no shape, only a clock — but the clock is read at amplitude $\Omega = W/\varepsilon$, which is what the scaling of [Eq. \(2.4\)](#) achieves. An oscillation of amplitude of order one is read at $\Omega \rightarrow \infty$, where the profile potential is dominated by its linear term and the orbit is a long ballistic arc with a brief rebound: P_0 grows like $2\sqrt{2\Omega}$ while D_0 grows only logarithmically, so by [Eq. \(3.6\)](#)

$$P \rightarrow 2\sqrt{2W}, \quad \frac{P_{\text{dip}}}{P} = \frac{D_0(W/\varepsilon)}{P_0(W/\varepsilon)} \rightarrow 0, \quad (3.7)$$

the period tending to a limit of order one and the dip fraction to zero. For the exponential profile $m_\theta = e^{-\xi}$ both are explicit, $D_0/P_0 = \log(4\Omega/c)/(2\Omega)(1 + o(1))$, so that

$$\frac{P_{\text{dip}}}{P} = \frac{\varepsilon}{2W} \log \frac{4W}{c\varepsilon} (1 + o(1)). \quad (3.8)$$

Had z been measured in layer thicknesses, Ω would have been W itself, D_0/P_0 a fixed number at fixed amplitude, and nothing would have concentrated. This is one of the two places the scaling does work; Lemma 3.4 is the other.

Lemma 3.4 (The impulse of a pass is its entry speed). *Let a dip occupy $[\alpha, \beta]$ with $z(\alpha) = z(\beta) = 0$, and write $I = \int_{\alpha}^{\beta} m_{\theta}^{\varepsilon}(z) d\tau$ for its impulse and $P_{\text{dip}} = \beta - \alpha$. Then, exactly,*

$$cI = (v(\beta) - v(\alpha)) + P_{\text{dip}} + \int_{\alpha}^{\beta} \kappa v d\tau. \quad (3.9)$$

At frozen (c, θ) the dip is elastic — $\Phi_{c, \theta}(0) = 0$, so Eq. (2.6) gives $\frac{1}{2}v^2 = W$ at both crossings and $v(\beta) = |v(\alpha)| = \sqrt{2W}$ — and then

$$\left| I - \frac{2\sqrt{2W}}{c} \right| \leq \frac{(1 + \kappa_{\max} v_{\max}) P_{\text{dip}}}{c_{\min}}. \quad (3.10)$$

Proof. Integrate Eq. (2.2) over $[\alpha, \beta]$ and rearrange: that is Eq. (3.9), with no approximation of any kind. For Eq. (3.10), substitute $v(\beta) - v(\alpha) = 2\sqrt{2W}$ and bound the last two terms of Eq. (3.9) by $(1 + \kappa_{\max} v_{\max}) P_{\text{dip}}$, then divide by $c \geq c_{\min}$. $\square$

Together Eqs. (3.7) and (3.10) are concentration stated with constants: the pass occupies a vanishing fraction of its oscillation and nonetheless delivers an impulse of order one, fixed by the speed at which the oscillation entered — the duration goes to zero and the effect does not. In the alternative scaling the same calculation returns $I = O(\sqrt{\varepsilon})$ and a pass leaves no trace.

3.3 The excursion decomposition, and the one threshold

The identities so far are either exact along the trajectory (Lemmas 3.1 and 3.4) or statements about the frozen, undamped system (Lemmas 3.2, 3.3 and 2.8). The proof must not confuse the two, so we fix the decomposition of $[0, T]$ by the trajectory's own crossings of the section and nothing else.

Definition 3.5 (Passes, arcs, and the fast part). Fix $W_0 \in (0, W_{\max}]$ and put $v_0 = \sqrt{2W_0}$. A *pass* is a maximal subinterval of $[0, T]$ on which $z \leq 0$; an *arc* is a maximal subinterval on which $z \geq 0$. A pass is a *bubble* if the amplitude at the time it begins is at least W_0 , and an arc is *ballistic* if the amplitude at the time it begins is at least W_0 . The *fast part* is the union of the remaining passes and arcs — every interval on which the amplitude is below W_0 — and on it

$$|v| \leq v_1 := \sqrt{2(W_0 + |\Phi_{\min}|)}, \quad \Phi_{\min} = \min_z \Phi_{c, \theta}(z) = \varepsilon \Phi_{c, \theta}^0(\xi_*) = O(\varepsilon), \quad (3.11)$$

by Eq. (2.6) and Lemma 2.8. Nothing here refers to a periodic orbit: a pass is an interval between two crossings of a section, and that is all.

What the frozen system supplies is a *comparison*, used in one place only — to bound how long a pass can last, which is Lemma 3.6. Everything else descends from Eqs. (3.9), (3.11) and (5.1), all three of which hold along the trajectory as it is.

Lemma 3.6 (A pass is no longer than the frozen pass beneath it). *Let a pass occupy $[\alpha, \beta]$ with amplitude $W = W(\alpha) \geq W_0$ and control $c = c(\alpha)$, and set $\delta = (\kappa_{\max} v_{\max}^2 + M_* \dot{c}_{\max}) P_{\text{dip}}$, which bounds the drift of the amplitude across the pass by Eq. (3.1). Then*

$$P_{\text{dip}} \leq \sqrt{\varepsilon} \sup \left\{ D_0(\Omega', c', \theta) : \Omega' \geq \frac{W - \delta}{\varepsilon}, |c' - c| \leq \dot{c}_{\max} P_{\text{dip}} \right\}. \quad (3.12)$$

Proof. On $[\alpha, \beta]$, integrating Eq. (3.1) gives $W(\tau) \geq W - \delta$, and Assumption 2.5 gives $|c(\tau) - c| \leq \dot{c}_{\max} P_{\text{dip}}$. By Eq. (2.6), $\frac{1}{2}v(\tau)^2 = W(\tau) - \Phi_{c(\tau), \theta}(z(\tau))$, so at each altitude the speed is at least that of the frozen system at the same altitude with amplitude $W - \delta$ and the least favourable admissible c' . Since $P_{\text{dip}} = 2 \int_{z_-}^0 dz/|v|$ and the integrand decreases as $|v|$ grows, substituting that lower bound can only increase the integral, giving Eq. (3.12) through Eq. (3.6). The bound is implicit in P_{dip} through δ and closes by continuity: δ is of order P_{dip} , so δ/ε is of order $\log(1/\varepsilon)$ against W/ε of order $1/\varepsilon$, and the shift in Ω' is relatively negligible. $\square$

Equation (3.12) is the only inequality here that compares the trajectory with a frozen orbit, and it is affordable because a pass is short: over an interval of length P_{dip} the amplitude and the control cannot move by more than $O(P_{\text{dip}})$, so the comparison is local and self-improving and no adiabatic theorem over the horizon is needed. A construction asking the trajectory to track a closed orbit over many periods would need one.

With [Definition 3.5](#) fixed, one number sets both errors. Let

$$\vartheta_1 = \sqrt{\varepsilon} \sup\{D_0(\Omega, c, \theta) : \Omega \geq \frac{W_0}{2\varepsilon}, c \in [c_{\min}, c_{\max}], \theta \in \Theta\}, \quad (3.13)$$

the longest pass any bubble can have, the factor 2 absorbing the shift of [Eq. \(3.12\)](#) for all small ε . There is no second threshold: the error of averaging the fast part is set by v_1 of [Eq. \(3.11\)](#) directly, as [Lemma 4.6](#) shows, so W_0 alone governs both.

Lemma 3.7 (How small the threshold makes each error). *Under [Assumptions 2.1](#) and [2.6](#) the supremum in [Eq. \(3.13\)](#) is finite, and as $\varepsilon \rightarrow 0$ at fixed W_0 ,*

$$\vartheta_1 \rightarrow 0, \quad v_1 \rightarrow \sqrt{2W_0}.$$

For the exponential profile the supremum is attained at the left endpoint once ε is small, and

$$\vartheta_1 = \frac{2\varepsilon}{\sqrt{W_0}} \log \frac{2W_0}{c\varepsilon} (1 + o(1)).$$

Proof. Finiteness is [Eq. \(3.6\)](#) with Ω confined by [Eq. \(3.5\)](#) and [Assumption 2.6](#). By [Eq. \(3.7\)](#) and $P_0(\Omega) \rightarrow \infty$ we have $\sqrt{\varepsilon}D_0(\Omega) \rightarrow 0$ uniformly for $\Omega \geq W_0/(2\varepsilon)$, which gives the first limit; for the exponential profile D_0 is eventually decreasing in Ω by [Eq. \(3.8\)](#), so the supremum is its value at $\Omega = W_0/(2\varepsilon)$, which is $\sqrt{\varepsilon} \cdot 2 \log(4\Omega/c)/\sqrt{2\Omega}$ evaluated there. The second limit is [Eq. \(3.11\)](#) with $\Phi_{\min} = O(\varepsilon)$. No monotonicity of D_0 in Ω is assumed, which is why [Eq. \(3.13\)](#) is a supremum. $\square$

Remark 3.8 (What the threshold trades). The two errors move oppositely in W_0 , and choosing between them is [Remark 4.4](#). Raising W_0 admits fewer bubbles and shortens each pass, so ϑ_1 falls; but it leaves more of the motion to be averaged at a larger speed excursion, so v_1 rises like $\sqrt{2W_0}$. Since ϑ_1 carries a factor $\varepsilon \log(1/\varepsilon)$ and v_1 does not, the best threshold shrinks with ε . The count of bubbles moves with it: by [Eq. \(3.10\)](#) each has impulse at least $2\sqrt{2W_0}/c_{\max}$, so [Assumption 2.4](#) admits at most

$$N_* = \frac{c_{\max}M}{2\sqrt{2W_0}} \quad (3.14)$$

of them — a bound free of ε at fixed W_0 , obtained from an exact integral identity and a budget and from nothing else, and which grows as W_0 is lowered to buy a smaller residual. This is the profile count the classical argument obtains only by contradiction and without a constant.

3.4 The bubble

Definition 3.9 (Pass map). For a bubble, let (v^-, s^-) be its data where it last crosses $z = 0$ downward and (v^+, s^+) where it next crosses upward, computed from [Eqs. \(2.1\)](#) to [\(2.2\)](#) together with the *inner* slow equation $\dot{s} = m_\theta^\varepsilon(z) \Gamma(s, c, \theta)$ — that is, [Eq. \(2.3\)](#) with the drift F dropped, whose contribution over a dip is at most $F_{\max}P_{\text{dip}}$. The *pass map* is

$$J_\theta(v^-, s^-) = \{(v^+, s^+) : \text{measurable } c(\cdot) \text{ on the dip}\},$$

at the trajectory's own θ ; we drop the subscript where it is fixed.

Lemma 3.10 (The bubble is nearly elastic, and bounded in closed form). *Let $v^- \leq -v_0 < 0$ with $v_0 = \sqrt{2W_0}$, which every bubble satisfies because $\frac{1}{2}(v^-)^2 = W \geq W_0$. Along every orbit of [Definition 3.9](#):*

(a) *at frozen (c, θ) the dip is elastic: since $\Phi_{c,\theta}(0) = 0$, [Eq. \(2.6\)](#) gives $\frac{1}{2}(v^\pm)^2 = W$ at both crossings, so $v^+ = |v^-|$ exactly;*

(b) *in general W drifts across the dip by at most $(\kappa_{\max}v_{\max}^2 + M_*\dot{c}_{\max})P_{\text{dip}}$ by [Eq. \(3.1\)](#), so*

$$|v^+ - |v^-|| \leq \frac{(\kappa_{\max}v_{\max}^2 + M_*\dot{c}_{\max})P_{\text{dip}}}{v_0};$$

(c) *the dip reaches no lower than the z_- with $\Phi_{c,\theta}(z_-) = W$, which exists and is unique by [Lemma 2.8](#), and $|z_-| \rightarrow 0$ with ε ;*

(d) *its impulse I obeys [Eq. \(3.10\)](#), and $|s^+ - s^-| \leq \Gamma_{\max}I$.*

Proof. (a) On the dip the orbit is the level set of [Eq. \(2.6\)](#), so $\frac{1}{2}v^2 = W - \Phi_{c,\theta}(z)$ throughout; at $z = 0$ the potential vanishes because $M_\theta^\varepsilon(0) = 0$, giving $\frac{1}{2}(v^\pm)^2 = W$ with the same W at both crossings. (b) Away from frozen data W is no longer constant, and [Eq. \(3.1\)](#) bounds its rate by $\kappa_{\max}v_{\max}^2 + M_*\dot{c}_{\max}$ with M_* from [Eq. \(2.7\)](#); integrating over the dip and dividing by $|d(\frac{1}{2}v^2)/dv| = |v| \geq v_0$ gives the displayed bound. (c) is [Lemma 2.8](#), and $z_- = \varepsilon\xi_-(\Omega)$ with $|\xi_-| = O(\log \Omega)$ by [Lemma 3.3](#). (d) is [Lemma 3.4](#) together with $s^+ - s^- = \int_{\text{dip}} m_\theta^\varepsilon \Gamma d\tau$. $\square$

Lemma 3.11 (The pass map, and how an error crosses it). *Let two entry data for the same (c, θ) differ by $(\delta v, \delta s)$, with both $|v^-| \geq v_0$. Then the exits differ by $(\delta v^+, \delta s^+)$ with*

$$|\delta v^+| = |\delta v|, \quad |\delta s^+| \leq e^{\kappa_\Gamma I} |\delta s| + \frac{2\Gamma_{\max}}{c_{\min}} |\delta v|, \quad (3.15)$$

where I is the impulse of the dip. Consequently J is Lipschitz in the Hausdorff distance, and — what the proof needs — the factor multiplying the slow error is exponential in that dip's own impulse.

Proof. Parametrise the dip by the delivered impulse $\mu = \int m_\theta^\varepsilon d\tau \in [0, I]$, in which $ds/d\mu = \Gamma(s, c, \theta)$ — Lipschitz in s with constant κ_Γ and involving neither z nor v — so Gronwall over the horizon I gives $e^{\kappa_\Gamma I}$ on δs . The speed is an isometry: $v^+ = |v^-|$ at frozen data by Lemma 3.10(a), so $|\delta v^+| = |\delta v|$ with no amplification and no Gronwall on v . What remains is the exit's dependence on δv through the horizon: by Eq. (3.10), $|\partial I / \partial v^-| \leq 2/c_{\min}$ up to $O(P_{\text{dip}})$ and $|\partial s^+ / \partial I| \leq \Gamma_{\max}$, which is the second term of Eq. (3.15). Suprema over (c, θ) give the Hausdorff statement. $\square$

The form of Eq. (3.15) is what matters downstream. The slow error is amplified by $e^{\kappa_\Gamma I}$, which tends to 1 with the pass's own impulse, so the product over any collection of bubbles is $e^{\kappa_\Gamma \Sigma I}$ and Assumption 2.4 bounds the exponent whatever their number; the coupling from δv enters *additively* rather than as a factor, and the speed error is not amplified at all, so the two cannot compound into a product over the passes.

4 The limit system, and the theorem

Definition 4.1 (The limit system). Fix $W_0 \in (0, W_{\max}]$ and put $v_0 = \sqrt{2W_0}$. The limit system $\mathcal{H}$ has state (z, v, s) on the arcs and (s) on the glide, and three modes.

Arc (flow). $\dot{z} = v$, $\dot{v} = -1 - \kappa v$, $\dot{s} = F(s, c, \theta)$, while $z \geq 0$: the ballistic arc, on which the air is not felt.

Bubble (jump). At $z = 0$ with $v \leq -v_0$ the state is replaced by $(0, v^+, s^+)$ with $(v^+, s^+) \in J(v^-, s^-)$, consuming no time.

Glide (averaged flow). On the fast part of Definition 3.5, where the amplitude is below W_0 : $\dot{s} = F(s, c, \theta) + \frac{1}{c} \Gamma(s, c, \theta)$, by Eq. (3.3), and z is not carried.

$\mathcal{R}_\mathcal{H}(t) \subseteq \mathbb{R}^{2+d}$ is the set of states reached at time t from $\mathcal{X}_0$ over all admissible c , all $\theta \in \Theta$ and all selections from J , with the convention that a state on the glide is carried as $(0, 0, s)$ — the glide mode does not propagate (z, v) , and Eq. (4.1) says what that costs. We write $\mathcal{R}_\mathcal{H}(E) = \bigcup_{t \in E} \mathcal{R}_\mathcal{H}(t)$.

That convention has a price, since $\mathcal{R}_\varepsilon(T)$ lives in $\mathbb{R}^{2+d}$ while the glide mode carries only s . Let $\Lambda = \{(z, v) : \frac{1}{2}v^2 + \Phi_{c,\theta}(z) \leq W_0\}$ be the fast states the fast part can occupy, which by Eq. (3.11) satisfies

$$\Lambda \subseteq \{|v| \leq v_1\} \times \{|z| \leq z_1\}, \quad z_1 = \sup\{|z| : \Phi_{c,\theta}(z) \leq W_0\}, \quad (4.1)$$

a set that shrinks with W_0 and with ε . A state the glide mode holds as s is to be read as the whole fibre $\Lambda \times \{s\}$ above it, and that is the lift Eq. (4.3) applies.

Note what the glide mode does *not* carry: by Eq. (3.3) the averaged slow field is independent of the amplitude, so no action coordinate is needed. The amplitude is an output of the reduced description. Note also that $\mathcal{H}$ is free of ε : the arcs are ballistic, the glide is averaged, and the only trace of the layer is the pass map, which Eq. (3.10) fixes to leading order in terms of the entry speed alone.

We measure the residual in $|(z, v, s)| = |z| + |v| + |s|$ and write $\mathbb{B}(\varrho)$ for the corresponding ball, $\oplus$ for the Minkowski sum. The constants are

$$C_f = F_{\max} + m_+ \Gamma_{\max}, \quad \kappa_0 = \kappa_F + \frac{\kappa_\Gamma}{c_{\min}}, \quad G = e^{\kappa_0 T + \kappa_\Gamma M}, \quad C_A = (\Gamma_{\max} + \kappa_\Gamma C_f T) K, \quad K = \frac{2}{c_{\min}} + \left(\frac{\dot{c}_{\max}}{c_{\min}^2} + \frac{\kappa_{\max}}{c_{\min}} \right) T, \quad (4.2)$$

with $m_+ = \sup_\Theta m_\theta(0)$ and M from Assumption 2.4. Every one is free of ε , which is the one discipline the proof keeps: inside the band, where the density is unbounded as $\varepsilon \rightarrow 0$, it is bounded by its *impulse* and never pointwise, and Assumption 2.4 bounds every impulse that occurs. The pointwise bound m_+ is used only above the section, where Assumption 2.1 makes it uniform.

Theorem 4.2 (Outer inclusion with an explicit residual). *Assume Assumptions 2.1 to 2.6 and fix $W_0 \in (0, W_{\max}]$. Let ϑ_1 be the threshold Eq. (3.13), v_1 the speed bound Eq. (3.11), N_* the count Eq. (3.14), Λ the fibre Eq. (4.1),*

and A_ε the impulse felt on the ballistic arcs, for which $A_\varepsilon \leq \varepsilon M_\infty c_{\max} M / (2W_0)$. Then

$$\mathcal{R}_\varepsilon(T) \subseteq \underbrace{\left[\mathcal{R}_\mathcal{H}([T - \varrho, T + \varrho]) + \Lambda \times \{0\} \right]}_{\text{skeleton, bubbles and glide}} \oplus \underbrace{\mathbb{B}(\varrho)}_{\text{residual}}, \quad (4.3)$$

where the sum with $\Lambda \times \{0\}$ lifts the glide mode's slow states to the fast fibre they stand for, and

$$\varrho = G \left(\underbrace{F_{\max} N_* \vartheta_1}_{\text{the drift over the passes}} + \underbrace{C_A v_1}_{\text{averaging the fast part}} + \underbrace{\frac{2\Gamma_{\max}}{c_{\min}} N_* c_{\max} A_\varepsilon}_{\text{the air the arcs discard}} \right). \quad (4.4)$$

All constants are those of [Assumptions 2.1 to 2.6](#) through [Eq. \(4.2\)](#); none depends on the control, the selection, the initial point or ε . Taking the union over $\theta \in \Theta$ on both sides gives the statement for the whole admissible family.

Three remarks. The inclusion is over a short interval of times, of half-width the residual, because a pass of positive duration is compared with an instantaneous jump. The lift by Λ is not slack dressed as rigour: the averaged mode genuinely does not know where in the fast fibre the motion sits, and Λ is exactly the fibre the fast part can occupy, shrinking with W_0 alongside the residual. And each term of [Eq. \(4.4\)](#) is the error of one approximation, traced to one estimate below, so the theorem is modular and ϱ is a number rather than a rate; what the threshold does as $\varepsilon \rightarrow 0$ is [Remark 4.4](#).

Remark 4.3 (What the theorem is, and is not, as a decomposition). [Theorem 4.2](#) is not a profile decomposition and does not establish one: it compares two trajectories, and profiles, their scales and their orthogonality appear nowhere in its statement or proof. Two things are nevertheless true of it, and the distinction is the whole methodological claim. It is the quantitative estimate any decomposition must reproduce — such a decomposition would have to yield [Eq. \(4.4\)](#), with [Eq. \(3.14\)](#) as its profile count. And it shows the concentration structure can be reached *without* that machinery, by an exact integral identity and a budget. Whether the machinery buys anything the elementary route does not is [objective 1](#), and it is a real question.

Remark 4.4 (The threshold at its best, and the rate that follows). By [Lemma 3.7](#) the first term of [Eq. \(4.4\)](#) falls like $\varepsilon \log(1/\varepsilon)/W_0$ and the second rises like $\sqrt{W_0}$, while the third falls like $\varepsilon/W_0^{3/2}$, so there is a best threshold and it shrinks with ε . Balancing the second against the third gives $W_0 \asymp \sqrt{\varepsilon}$ and

$$\varrho = O(\varepsilon^{1/4}), \quad (4.5)$$

at which threshold the first term is $O(\sqrt{\varepsilon} \log(1/\varepsilon))$ and is not binding — so the logarithm of [Eq. \(3.8\)](#) does not reach the rate. We record [Eq. \(4.5\)](#) for the order of magnitude and claim nothing about its sharpness: it is what balancing three crude bounds gives. Which of the three is the real obstruction is a quantitative question we would want settled early, and the third — the air felt on the arcs, which is an artefact of cutting a pass at $z = 0$ rather than at the edge of the band — is the one we would expect to improve first.

Remark 4.5 (At a fixed ε , and what the threshold then costs). Nothing above requires ε to be small: [Eq. \(4.4\)](#) holds at every ε and every W_0 , and at a fixed physical ε it is evaluated rather than estimated. What it then says is that the residual is small only if ϑ_1 and v_1 both are, and [Section 7](#) shows that for entry they are not: no threshold makes the passes brief and the averaged excursion small together. The remedy is not a new error term but a narrower description. The motion near the threshold, which neither treatment serves, may be *propagated exactly* — carrying the fast pair (z, v) and the unapproximated field [Eqs. \(2.1\) to \(2.3\)](#) — and this adds nothing to ϱ because nothing is approximated, and removes the lift Λ over those windows because (z, v) is then carried. What it costs is dimension and amplification: the reduced problem carries $2 + d$ coordinates instead of d there, and G acquires the factor $e^{\kappa_0 T_E}$ for the total duration T_E of those windows, which [Eq. \(3.4\)](#) bounds by $2c_{\max} M$ once the windows are taken to be the intervals on which $m_\theta^\varepsilon \geq 1/(2c_{\max})$. So the threshold trades sharpness against dimension, not sharpness against sharpness.

Lemma 4.6 (Averaging the fast part, with its constant). *Let s solve [Eq. \(2.3\)](#) on a subinterval of the fast part and let $\bar{s}$ solve $\dot{\bar{s}} = F(\bar{s}, c, \theta) + \frac{1}{c} \Gamma(\bar{s}, c, \theta)$ with the same initial value. Then*

$$|s - \bar{s}| \leq G C_A v_1$$

with G, C_A as in [Eq. \(4.2\)](#) and v_1 as in [Eq. \(3.11\)](#); the factor G is the amplification over the horizon, and it is the same G that [Eq. \(4.4\)](#) collects once in front of every term, so C_A carries none of it.

Proof. Put $e = s - \bar{s}$. Subtracting, $\dot{e} = [F(s) - F(\bar{s})] + m_\theta^\varepsilon [\Gamma(s) - \Gamma(\bar{s})] + (m_\theta^\varepsilon - \frac{1}{c}) \Gamma(\bar{s})$, so $|\dot{e}| \leq (\kappa_F + \kappa_\Gamma m_\theta^\varepsilon) |e| + |(m_\theta^\varepsilon - \frac{1}{c}) \Gamma(\bar{s})|$.

Everything turns on the primitive $R(\tau) = \int_0^\tau (m_\theta^\varepsilon - \frac{1}{c})$, and it is bounded *from the equation of motion*, with no appeal to periodicity and none to a frozen orbit. By Eq. (2.2), $c m_\theta^\varepsilon - 1 = \dot{v} + \kappa v$, so $m_\theta^\varepsilon - \frac{1}{c} = (\dot{v} + \kappa v)/c$ and, integrating by parts in the first term,

$$R(\tau) = \left[\frac{v}{c} \right]_0^\tau + \int_0^\tau \frac{v \dot{c}}{c^2} d\tau + \int_0^\tau \frac{\kappa v}{c} d\tau, \quad (4.6)$$

an identity, exact for every control and every selection. On the fast part $|v| \leq v_1$ by Eq. (3.11), so $|R| \leq v_1(2/c_{\min} + (\dot{c}_{\max}/c_{\min}^2 + \kappa_{\max}/c_{\min})T) = v_1 K$. Note what has replaced what: the zero-mean cancellation that periodicity would have supplied is here the statement that v is *small* on the fast part, which is a property of the trajectory and not of an orbit.

Integrating the inhomogeneous term by parts against R ,

$$\left| \int_0^\tau (m_\theta^\varepsilon - \frac{1}{c}) \Gamma(\bar{s}) \right| \leq |R| \Gamma_{\max} + |R| \int_0^\tau \left| \frac{d}{d\tau} \Gamma(\bar{s}) \right| \leq v_1 K (\Gamma_{\max} + \kappa_\Gamma C_f T),$$

using $|\dot{\bar{s}}| \leq C_f$. Gronwall's inequality in integrated form multiplies this by $\exp \int_0^T (\kappa_F + \kappa_\Gamma m_\theta^\varepsilon) \leq e^{\kappa_F T + \kappa_\Gamma M} \leq G$, by Assumption 2.4 and not by a pointwise bound on the density, which gives the claim. The estimate does not accumulate over the successive excursions of the fast part: R is bounded over the whole horizon by Eq. (4.6), not excursion by excursion. $\square$

5 Proof of Theorem 4.2

Fix $\varepsilon \in (0, 1]$, a threshold $W_0 \in (0, W_{\max}]$, and an admissible trajectory $x_\varepsilon = (z, v, s)$ on $[0, T]$ with control $c(\cdot)$, selection $\theta \in \Theta$ and $x_\varepsilon(0) \in \mathcal{X}_0$.

Step 1: the budget

By Assumption 2.4, $\frac{d}{d\tau} \Theta_d(s) = \nabla \Theta_d \cdot F + m_\theta^\varepsilon \nabla \Theta_d \cdot \Gamma \leq -c_0 m_\theta^\varepsilon$, so

$$\int_0^T m_\theta^\varepsilon(z) d\tau \leq M, \quad (5.1)$$

for every control, every selection and every ε .

Step 2: the excursions, and the three counts

By Definition 3.5 the interval $[0, T]$ is the disjoint union of passes and arcs, cut at the trajectory's own crossings of $z = 0$, each classified by the amplitude where it begins. No periodicity is used and no orbit is asserted to close. Three counts follow, and each descends from Eq. (5.1) together with one exact identity.

How many bubbles. By Eq. (3.10) a bubble entered at amplitude $W \geq W_0$ delivers impulse at least $2\sqrt{2W_0}/c_{\max}$, up to the $O(P_{\text{dip}})$ of Eq. (3.10) which is absorbed for small ε . Since the impulses of disjoint passes sum to at most M by Eq. (5.1), there are at most the N_* of Eq. (3.14). This is Eq. (3.9) and a budget, and nothing else.

How long the passes last in total. By Lemma 3.6 each bubble has $P_{\text{dip}} \leq \vartheta_1$, so

$$\sum_{\text{bubbles}} P_{\text{dip}} \leq N_* \vartheta_1. \quad (5.2)$$

How much air the ballistic arcs discard. On an arc $z \geq 0$, so $m_\theta^\varepsilon(z) = m_\theta(z/\varepsilon) \leq m_+$, and by Lemma 3.3 the impulse of a single arc is $\sqrt{\varepsilon} \int_0^{\xi^+} m_\theta(\xi) d\xi/u \leq \varepsilon 2M_\infty/\sqrt{2W_0}$ for an arc entered above the threshold. Summing over at most N_* of them,

$$A_\varepsilon := \sum_{\text{ballistic arcs}} \int m_\theta^\varepsilon d\tau \leq \frac{\varepsilon M_\infty c_{\max} M}{2W_0}. \quad (5.3)$$

The number of passes grows as the threshold is lowered while each pass shortens like $\varepsilon \log(1/\varepsilon)$, and Eq. (5.2) says the product is controlled; neither it nor Eq. (5.3) is a compactness argument, and neither could be obtained from one.

Step 3: the three treatments exhaust the motion

By Definition 3.5 every point of $[0, T]$ lies in a bubble, a ballistic arc, or the fast part — precisely the three modes of Definition 4.1. Nothing is assumed: the classification is by the sign of z and by whether the amplitude starting the excursion reaches W_0 , both properties of the trajectory. What must be quantified is how good each treatment is, and that is Lemma 3.7.

Step 4: a bubble is a jump, up to the drift over its pass

Let a bubble occupy $[\alpha, \beta]$ with $z(\alpha) = z(\beta) = 0$. By [Definition 3.9](#) the exit data of the inner problem from $(v(\alpha), s(\alpha))$ differ from $(v(\beta), s(\beta))$ only through the one term dropped in [Eq. \(2.3\)](#), namely the drift F , whose integral over the pass is at most $F_{\max} P_{\text{dip}}$. Hence

$$(v(\beta), s(\beta)) \in J(v(\alpha), s(\alpha)) \oplus \mathbb{B}(F_{\max} P_{\text{dip}}), \quad (5.4)$$

and $z(\alpha) = z(\beta) = 0$ exactly, so the altitude contributes nothing. $F_{\max}$ appears and not C_f because the inner problem retains the density term of [Eq. \(2.3\)](#) — the only reason this step survives a density unbounded inside the band.

Step 5: the fast part is the averaged glide

This is [Lemma 4.6](#), giving $G C_A v_1$ — the G being the horizon amplification [Eq. \(4.4\)](#) collects in front of every term, counted here once and not again in Step 6. It is not an averaging theorem over a periodic orbit: no orbit is assumed to close, the control varies, the damping acts, and [Eq. \(4.6\)](#) replaces the periodic argument's zero-mean cancellation by the smallness of v where the motion is averaged. Averaging over *several* phases, whose frequencies may cross, is a different problem, is attempted nowhere here, and is [objective 3](#).

Step 6: assembling, and why the amplification is bounded

Construct $x_{\mathcal{H}}$ by following $\mathcal{H}$ with the same c and the same θ , jumping at each bubble with the point of J nearest to x_{ε} 's exit, flowing ballistically on each ballistic arc, and flowing with the averaged field on the fast part. Write δ for the discrepancy and track its two components separately, which is what [Eq. \(3.15\)](#) is for.

The slow error is amplified in two places, in both by a factor exponential in an impulse. On the arcs and the fast part both motions obey fields Lipschitz with constant κ_0 , giving $e^{\kappa_0 T}$ over the horizon. Across a bubble of impulse I , [Eq. \(3.15\)](#) gives $e^{\kappa_{\Gamma} I}$, so the product over all bubbles is

$$\prod_{\text{bubbles}} e^{\kappa_{\Gamma} I_k} = e^{\kappa_{\Gamma} \sum_k I_k} \leq e^{\kappa_{\Gamma} M}$$

by [Eq. \(5.1\)](#). Together the two are the G of [Eq. \(4.2\)](#). This is the second place the budget does work no compactness argument could: the amplification compounds over every pass, however many, and is bounded anyway, because each factor is exponential in that pass's *own* impulse and the impulses are budget-limited. The step turns on the form of [Eq. \(3.15\)](#) and not its size: a single Lipschitz constant for J would exceed 1 by a fixed amount, and its product over the passes would diverge as the threshold was lowered.

The speed error is not amplified at all, by [Eq. \(3.15\)](#): the bubble is an isometry in v . It is only *injected*, on the ballistic arcs, where x_{ε} feels the density that $x_{\mathcal{H}}$ does not, at a total cost of $c_{\max} A_{\varepsilon}$ by [Eq. \(5.3\)](#). It reaches the slow error through the additive coupling of [Eq. \(3.15\)](#), once per bubble, for a total of at most $(2\Gamma_{\max}/c_{\min}) N_* c_{\max} A_{\varepsilon}$ — the third term of [Eq. \(4.4\)](#), and the only one in which the number of passes appears.

Collecting [Eqs. \(5.2\)](#) and [\(5.4\)](#) for the bubbles, Step 5 for the fast part and [Eq. \(5.3\)](#) for the arcs,

$$|x_{\varepsilon}(T) - x_{\mathcal{H}}(\hat{T})| \leq G \left(F_{\max} \sum_{\text{bubbles}} P_{\text{dip}} + C_A v_1 + \frac{2\Gamma_{\max}}{c_{\min}} N_* c_{\max} A_{\varepsilon} \right),$$

which with [Eq. \(5.2\)](#) is the ϱ of [Eq. \(4.4\)](#), the comparison being made after both states are lifted to the fast fibre Λ of [Eq. \(4.1\)](#) wherever $x_{\mathcal{H}}$ is on the glide. The arrival time $\hat{T}$ differs from T because x_{ε} spends the passes where $x_{\mathcal{H}}$ spends no time, and by [Eq. \(5.2\)](#) that discrepancy is at most $N_* \vartheta_1$, which is at most $\varrho/(GF_{\max})$ and so at most ϱ whenever $GF_{\max} \geq 1$. Hence $\hat{T} \in [T - \varrho, T + \varrho]$ and [Eq. \(4.3\)](#) follows. $\square$

Remark 5.1 (Where each hypothesis was used). [Assumption 2.1](#) was used twice: in [Lemma 2.8](#), from which the shape of the well descends, and for the uniform bound m_+ on the density above the section. [Assumption 2.4](#) was used in Step 1, and everything quantitative descends from it: the pass count [Eq. \(3.14\)](#), the summed passes [Eq. \(5.2\)](#), the arc impulse [Eq. \(5.3\)](#), the time spent in appreciable air [Eq. \(3.4\)](#), the bounded amplification of Step 6, and the two places in [Lemma 4.6](#) where a pointwise bound on the density would have failed. [Assumption 2.5](#) was used twice, for the *a priori* amplitude bound [Eq. \(3.2\)](#) and for the $\dot{c}$ terms in [Lemma 3.6](#) and [Eq. \(4.6\)](#). [Assumption 2.2](#) supplies the constants and fixes the selection along a trajectory, without which [Eq. \(3.1\)](#) would not close ([Remark 2.3](#)); [Assumption 2.6](#) supplies the rest.

Two things must be kept apart, since the proof would be wrong if they were run together. The frozen, undamped statements — [Lemmas 3.2, 3.3](#) and [2.8](#) — only *define* P_0, D_0 and the averaged field, and enter the estimates in one place, [Lemma 3.6](#), over a single short pass. The statements holding along the trajectory — [Eqs. \(3.9\), \(3.11\), \(4.6\)](#) and [\(5.1\)](#) — carry everything else, and none assumes an orbit closes. There are five hypotheses and no sixth: that

the three treatments cover the motion is [Definition 3.5](#), a definition. No compactness argument and no measure appears above.

6 The weak formulation

[Theorem 4.2](#) concerns trajectories and its proof uses no measures. The weak formulation is nevertheless where the question belongs: it makes the solution set of a nonconvex family tractable at all, by rendering the dynamics a linear constraint, and it is where the loss of compactness is visible.

With $\mathcal{U} = [c_{\min}, c_{\max}]$, the *occupation measure* of a trajectory is $\mu_\varepsilon \in \mathcal{M}_+([0, T] \times \mathcal{X} \times \mathcal{U} \times \Theta)$ with $\int \varphi d\mu_\varepsilon = \int_0^T \varphi(t, x_\varepsilon(t), c(t), \theta) dt$, and μ_0, μ_T are the initial and terminal measures. The topology is weak-*; the base is a compact metric space by [Assumption 2.2](#), so it is metrizable and bounded sets are relatively compact. The dynamics are then linear: for every $\varphi \in C^1$,

$$\int \varphi(T, \cdot) d\mu_T - \int \varphi(0, \cdot) d\mu_0 = \int \left(\frac{\partial \varphi}{\partial t} + \nabla \varphi \cdot (v, cm_\theta^\varepsilon - 1 - \kappa v, F + m_\theta^\varepsilon \Gamma) \right) d\mu_\varepsilon, \quad (6.1)$$

the Liouville equation of the family, whose terminal measures' supports union to the reachable set at time T . This is the occupation-measure programme of Lasserre, Henrion and collaborators [4, 5], and an outer bound comes from its dual: a pair (Φ, Ψ) with Φ non-decreasing along the admissible motions and at least one on $\mathcal{X}_0$, and Ψ majorising Φ at time T , encloses the reachable set in $\{\Psi \geq 1\}$. Exhibiting such a pair is a proof whether or not it is ever evaluated numerically.

[Theorem 4.2](#) hands this machinery a different problem. After it the object to enclose is $\mathcal{R}_\mathcal{H}$: a few coordinates with a flow, one guard $z = 0$, a set-valued reset J , and a jump count bounded by [Eq. \(3.14\)](#) — exactly the hybrid form for which the relaxation is known [6], a measure on the guard carrying the mass between modes and Φ not decreasing along the flow or across the reset. Two things remain. [Theorem 4.2](#) compares trajectories and not measures, which is [objective 4](#); and nothing above makes $\mathcal{R}_\mathcal{H}$ tight, so what is lost in bounding it is a separate question from what is lost in the limit. Whether the limit is better taken in this topology, with the decomposition extracted from the measures, or at the level of trajectories as in [Section 5](#), is the open question of [Section 1.1](#); the two routes are not known to be equivalent.

7 What hypersonic entry adds

The application is hypersonic entry: the reachable set of an entry vehicle, for any vehicle and any corridor, with both entering only as data. It supplies the singular perturbation, the limit system and the concentrating events in one system, and it is not already covered. The exact method solves a Hamilton–Jacobi–Isaacs equation on a grid, capped in practice near four or five states [7], while a rigid-body entry vehicle carries thirteen and more once ablation and the aerothermoelastic modes are coupled [8]; set-propagation methods return rigorous enclosures at ten to twenty states [9], but of one instance — one vehicle, one box, one horizon — with no explicit dependence on the corridor or on the width of the aerodynamic uncertainty, which a statement uniform over a family requires.

7.1 The correspondence

Measure altitude in planetary radii above the equilibrium glide, so the density is $m_\theta^\varepsilon(z) = m_\theta(z/\varepsilon)$ with the profile read in scale heights, and let $\varepsilon = H_s/R_e$ be the thin-atmosphere ratio, whose square root is the ratio of the oscillation period about the glide to the slow time $\sqrt{R_e/g}$. Then [Eqs. \(2.1\) to \(2.2\)](#) are the vehicle's vertical motion: v is the rate of climb, c the vertical lift fraction $(L/D)\cos\sigma$ normalised so that $cm_\theta^\varepsilon = 1$ is the glide balance — lift against weight less centrifugal relief — and κ the phugoid damping, which does not scale with ε . The slow state s collects the energy, the range and the heading, F carries the gravitational terms and Γ the drag. The selection θ chooses the density profile and the aerodynamic coefficients from an enclosure, one choice per flight by [Assumption 2.2](#), while the control still ranges over a set at each instant — so [Eqs. \(2.1\) to \(2.3\)](#) are a differential inclusion in the control over a family indexed by the uncertainty [10], and the vehicle enters only through dimensionless ratios, each quantified over a range for a class and never evaluated at a design. The three regimes are the content of the model rather than an analogy: the minimum of the well is the equilibrium glide, at height of order ε by [Lemma 2.8](#); a small oscillation is the phugoid, of period $\sqrt{\varepsilon}$ and damping ratio of the same order, so it persists for the whole entry; and a large one is a skip, of period of order one, rebounding in a time of order $\varepsilon \log(1/\varepsilon)$ by [Lemma 3.3](#) and delivering an impulse of order one by [Lemma 3.4](#).

[Assumption 2.4](#) is the hypothesis that transfers exactly. The specific mechanical energy in the rotating frame, $\mathcal{E} = \frac{1}{2}V^2 + \Phi_g - \frac{1}{2}\omega_E^2 r^2 \cos^2\phi$, satisfies $\dot{\mathcal{E}} = -P$ with $P = DV/m \geq 0$ the drag power, because neither the lift nor the Coriolis force does work; taking $\Theta_d = \mathcal{E}$ gives $\nabla \Theta_d \cdot F = 0$ exactly and $\nabla \Theta_d \cdot \Gamma \leq -c_0$ with c_0 fixed by lower

bounds on the drag coefficient and the speed. The remaining hypotheses are conditions on the corridor: bounded away from circular speed, so that the gravity term normalises; bank bounded away from ninety degrees, giving $c \geq c_{\min} > 0$; a glide altitude existing for every admissible bank, which is the last clause of [Assumption 2.1](#); and bounded away from the poles, where the chart is singular.

7.2 Why the residual is not small at the physical ε

[Theorem 4.2](#) applies to entry, but its residual is not small, and the reason is quantitative. A pass entered at flight-path angle γ_e crosses a scale height in a time of order ε/γ_e on the slow time scale and levels out under a load of order γ_e^2/ε relative to gravity, so the entry angle enters through the ratio $\gamma_e/\sqrt{\varepsilon}$, as in the classical Chapman variables [11]. For the Earth $\sqrt{\varepsilon} \approx 0.034$, about two degrees, and entry angles are a few degrees. So in the language of [Eq. \(3.13\)](#), no threshold W_0 makes ϑ_1 and v_1 small together: the amplitude at which a pass becomes brief is the amplitude at which the excursion being averaged becomes large. [Remark 4.5](#) rather than [Eq. \(4.5\)](#) is what applies, and it costs dimension rather than a new error term.

8 Plan of work

[Theorem 4.2](#) is Theorem C for one model, and it is discharged. Four obligations remain, and we state each with the obstruction it has to overcome. The first and fourth are the analysis; the second and third carry the model to the application.

Objective 1. Theorem A: the profile decomposition, with a constant

We must prove the decomposition itself: that every sequence of trajectories with $\varepsilon_n \rightarrow 0$ admits, along a subsequence, a splitting into a skeleton, finitely many profiles obtained by rescaling the passes to their own widths, and a remainder small in a stated norm. The classical route iterates the dichotomy of the concentration-compactness principle [1] and, in the form that resolves each profile at its own scale, extracts a pairwise orthogonal family [2, 12]; orthogonality is free here, since the scale of a profile is the width of a pass and its centre the time the pass occurs. What we expect to add is the constant. In that theory the number of profiles above a given size is finite because a conserved quantity is, and the argument is by contradiction; here [Assumption 2.4](#) bounds the count *in advance* and explicitly, which is [Eq. \(3.14\)](#). The objective is met by the decomposition stated with that count, its remainder measured in the share of dissipation held by any window of fixed rescaled width — the norm in which such a remainder is small, and in which it is not small in the conserved quantity [2, p. 136].

Objective 2. The statement at a fixed ε , and a time-varying atmosphere

[Theorem 4.2](#) holds for entry with a residual [Section 7](#) shows is not small at the physical ε . We must prove what [Remark 4.5](#) describes: the motion near the threshold carried by the unapproximated field, the reduced problem acquiring the fast pair over a duration [Eq. \(3.4\)](#) bounds, and the residual gaining no new term. That enclosure is a propagation over a bounded time rather than an asymptotic estimate, so what has to be controlled is its growth, explicit in the interval widths. The same objective carries the generalisation the base theorem defers. By [Assumption 2.2](#) the selection is fixed along each trajectory, and [Remark 2.3](#) says why; the budget is indifferent to this, being a common Lyapunov functional, so what must be rebuilt is the fast half of the argument and not the slow. We must state a class of time-varying selections for which the amplitude law survives — bounded variation in altitude, monotonicity where it holds, which is what the physics supplies — and carry it through [Lemmas 3.1, 3.6, 3.10 and 3.11](#). This is a genuine enlargement of the enclosure and not a sharpening of its constants.

Objective 3. The remainder: motions that do not concentrate

A profile decomposition separates what concentrates from what does not; the second part is the remainder, and the entry problem has motions of that kind which Eqs. (2.1) to (2.3) does not carry. The attitude and the elastic modes oscillate for as long as there is air and nothing about them concentrates, so they are not profiles at all. The classical answer is averaging [13], which for *several* phases reaches no bound: it gives an order, holds only away from resonance while the attitude frequency rises from zero at every pass, and the results carrying a solution through one discard a set of initial conditions an outer statement cannot. We propose a Lyapunov functional in its place — to bound what these motions *do* rather than resolve what they are. They act on the translational motion only through the aerodynamic force, so the force is what must be enclosed, and the selection can carry that enclosure at no cost in dimension. Two facts are to be established. Where there is no air there is neither torque nor aerodynamic force, so the attitude cannot act on the trajectory at all. Where there is air, the damping rate goes as density times speed, while the energy identity makes the integral of density against the *cube* of the speed exactly the energy shed; dividing by the square of the largest admissible speed turns that equality into a lower bound on the integrated damping. The objective is met by a bound on the aerodynamic force error, explicit in the dimensionless groups of the class and uniform over the corridor.

Objective 4. The measure-level companion and the certificate

We must show that the occupation measures of Section 6 converge to measures satisfying the Liouville equation of the limit system, with the mass crossing each pass carried on the graph of J , uniformly over the admissible family, and exhibit the dual pair (Φ, Ψ) . This is what makes Theorem 4.2 a statement about the limit of the family rather than a comparison of two trajectories, and where weak-* compactness and concentration-compactness in the form resolving each profile at its own scale [1] are expected to be needed — for the structure, the constants coming from Assumption 2.4 as in objective 1.

8.1 Sequencing

Three phases rather than four parallel tracks. The first extends Theorem 4.2 to the planar entry problem, where the correspondence above is exact and Assumptions 2.1 and 2.4 are verified rather than assumed; the question of which of the three terms of Eq. (4.4) is the real obstruction belongs to it, and so does objective 1, since the count Eq. (3.14) is in hand and what remains is the decomposition it would organise. The second is objective 2, in which the threshold and the selection class are the two dials on one statement. The third is objective 3, to be reached early enough that a negative result is still recoverable, with objective 4 alongside it, needing only that the limit system be settled. If the Lyapunov route does not close, the retreat is a corridor on which the frequency bands are ordered, the crossings at the edges of each pass left to the enclosure of the bubble, which integrates the coupled field and needs no separation — weakening the residual without invalidating the decomposition.

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